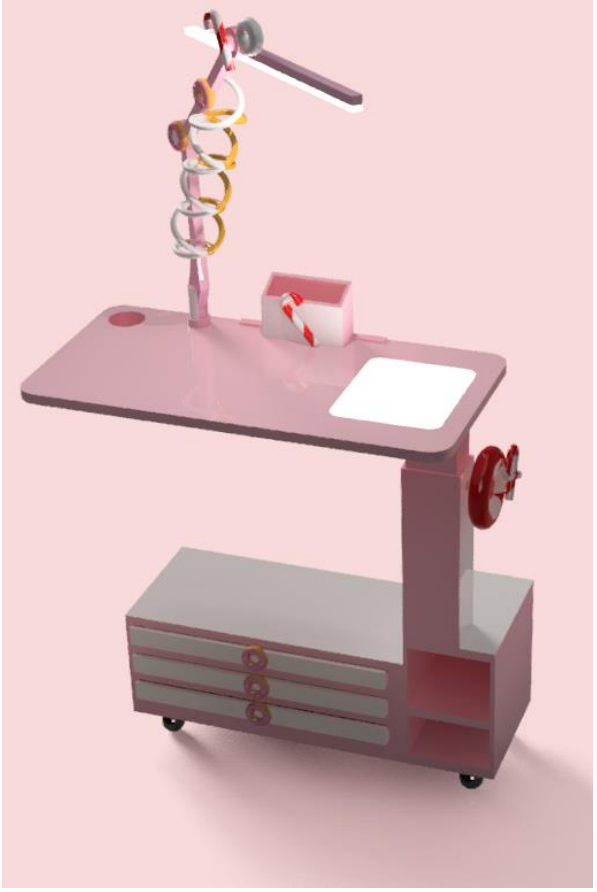
Personal CAD – Making Project

Candy Themed Scribe desk



PennState College of Engineering | ENGINEERING DESIGN

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[EDSGN 100, Fall 2025, 009, Dr. Yu Tianliang]

[DEC/2025]

PennState College of Engineering | ENGINEERING DESIGN

1. <u>Images</u>:

<u>1.1 Initial Concept</u>

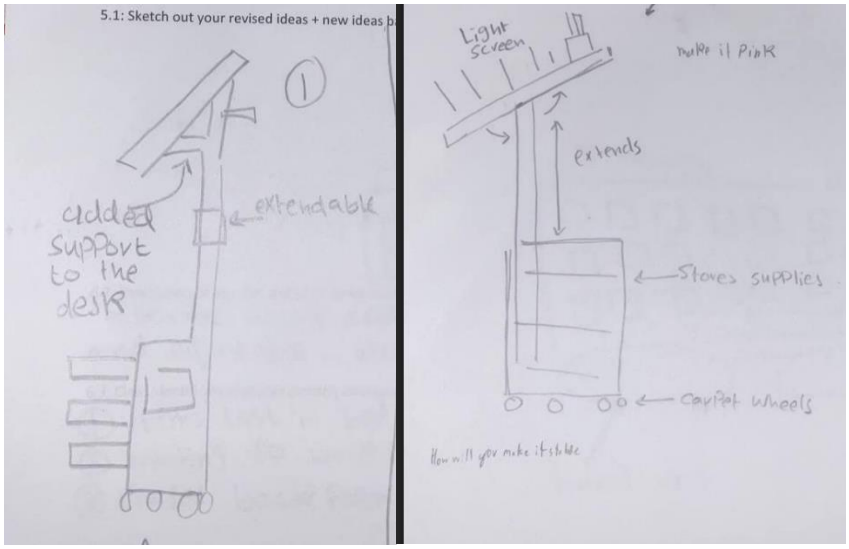

Figure 1.1.1 Initial concept sketches

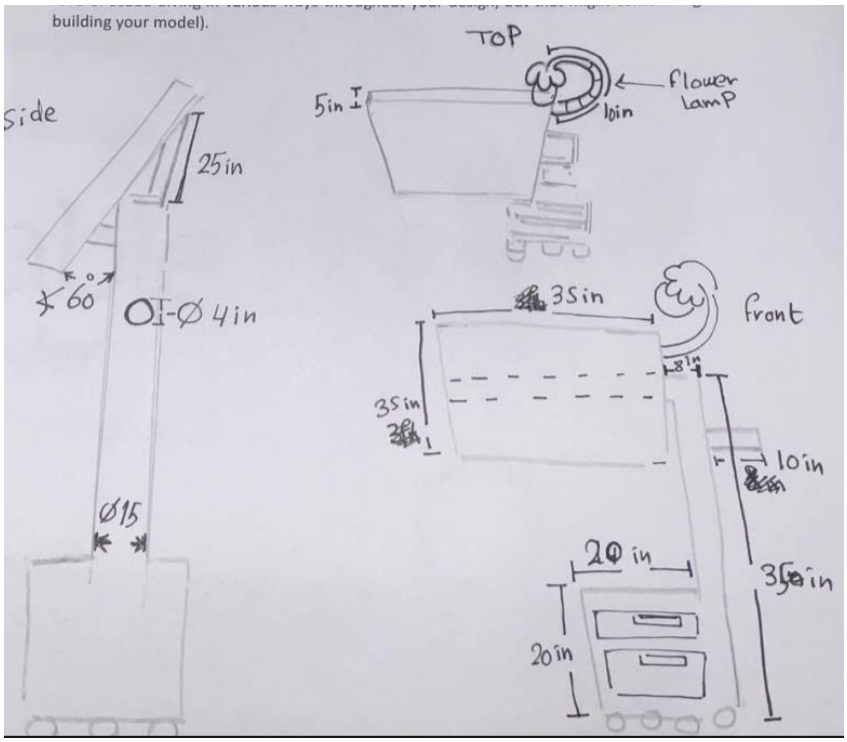

Figure 1.1.2 Initial concept dimensions

1.2 Fusion Parts and Features:

Figure 1.2.1 Desk body: with wheels, storage, a hook, stationary space and a cup holder

Figure 1.2.2 Donut shaped decoration used for drawer knobs and lamp decorations

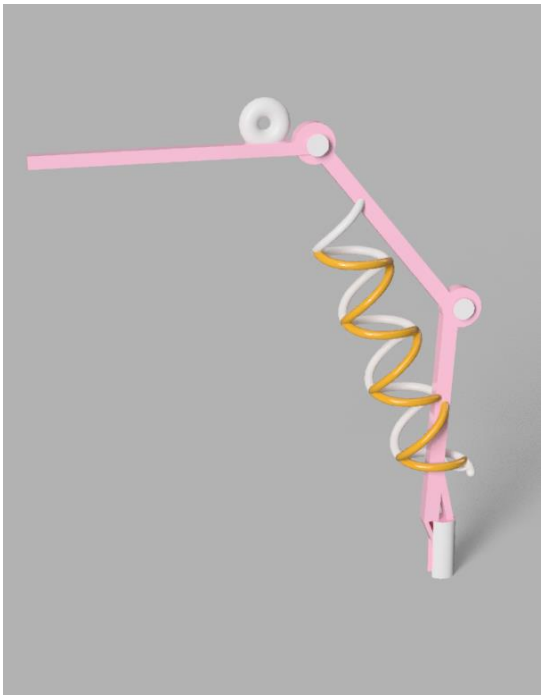

Figure 1.2.3 Lamb body with candy-like decoration

Figure 1.2.4 Drawer Basic body

Figure 1.2.5 candy cane shaped decoration used on multiple objects

Figure 1.2.6 Wheel handle for manual height adjustment

1.3 Drawing

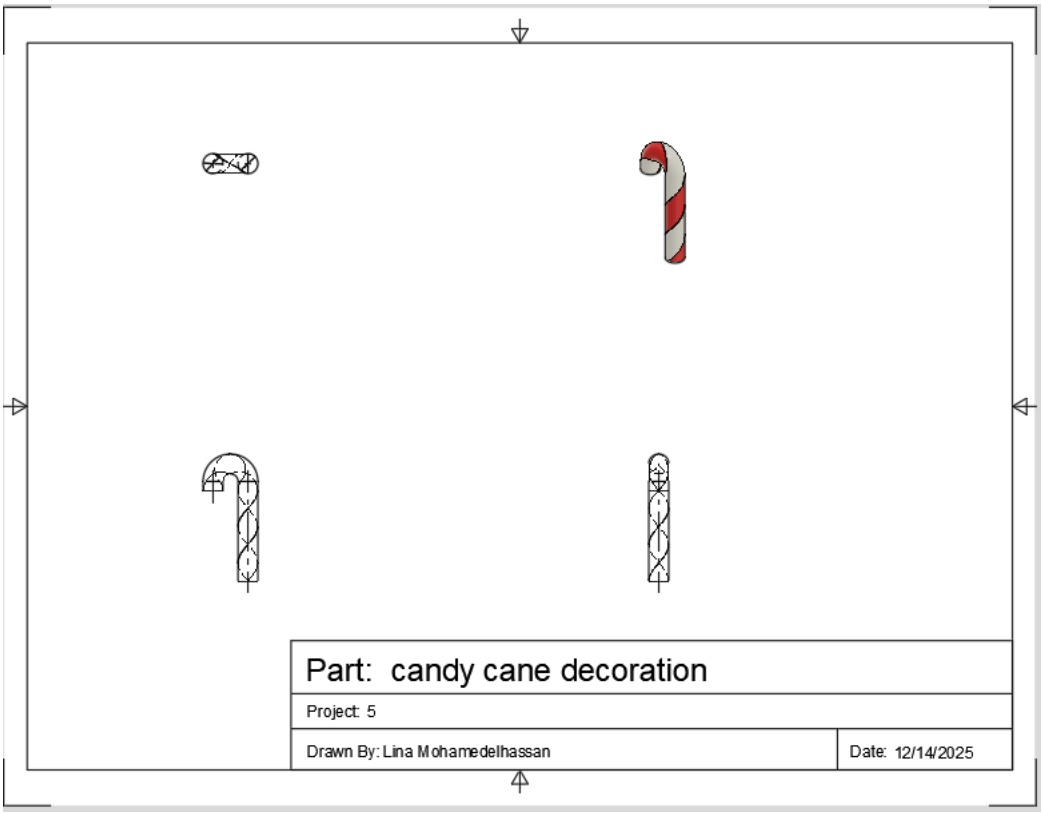

Figure 1.3. Dimensioned drawing of candy cane decorative component completed in Fusion

1.4 Final Assembly

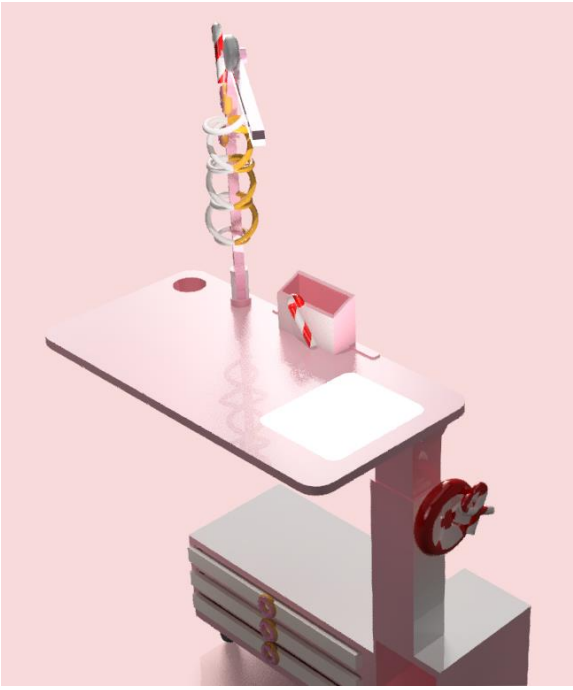

Figure 1.4.1 Assembled Product Right side view

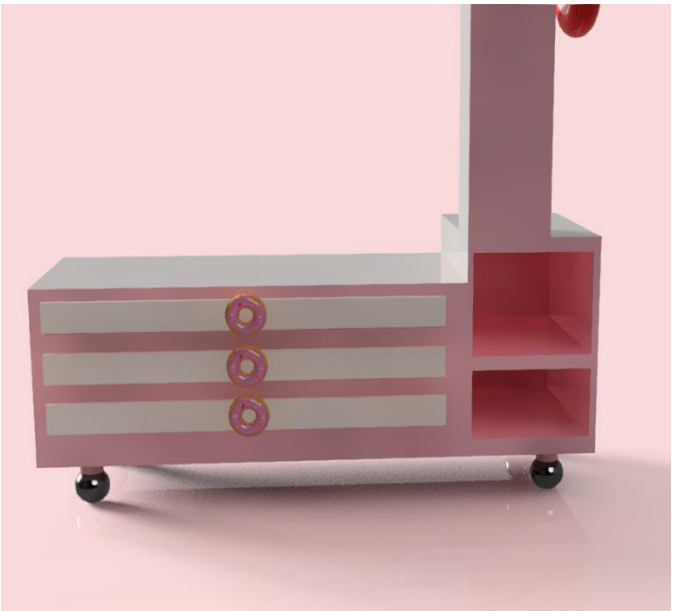

Figure 1.4.2 Assembled Product Bottom storage view

Figure 1.4.3 Assembled Product Back view

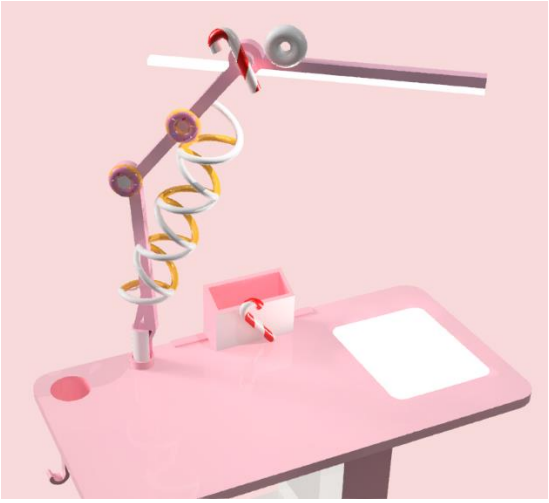

Figure 1.4.4 Assembled Product Top view

2. CAD Features

1.Mirror
2.Fillet
3.Draft
4.Champer
5.Sweep
6.Revolve
7.Extrude
8.Scale
9.Split body
10.Combine

I wanted to explore making even more candy themed objects and decorations, but I realized it will limit the functionality. So, I condensed the decoration into a few candy-themed pieces while maintaining a good working space to maintain the functionality of the desk. The main difference between the initial concept and the final designs is the switch of themes from plant themed to candy themed. This has led me to adjust the decoration, different parts designs and the general color scheme for the project.

3. Reflection

I learned the different parts used in building desks. I also learned about the different mechanisms used in extendable desks and the table-top tilting mechanisms. I did research on motor-based vs manually extendable desks. I implemented a light box on the desk: used for drawing. I had to learn about the different sizes and how they function without limiting the desks space. I learned about different wheels used in various spaces such as carpet wheels vs regular office chair wheels.

I would design a headphones' stand on top of the desk along and add extra decorations while still having enough workspace. I will also change the lamp design and change the orientation of the light box to make it tilt to about 30 ~ 45 degrees. Those changes would increase the aesthetics of the desk while maximizing efficiency in space and functionality. I would also add a phone/iPad stand on top of the workspace. All of those changes would require a change of dimensions.

4. Resources

[1] https://www.youtube.com/watch?v=Y1LgVlVbtAY
[2]https://www.youtube.com/watch?v=CR_CXkQAl5M